

Data Analysis Expression(DAX)

Second Assignment

Tasks

DAX Calculations

Calculated Columns

- Create a Calculated Column for 'Category Type': Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.

1 Category Type = 'List of Order'[Category]&" "&'List of Order'[Sub-Category]						
Amount	Profit	Quantity	Category	Sub-Category	Category Type	
561	212	3	Clothing	Saree	Clothing Saree	
119	-5	8	Clothing	Saree	Clothing Saree	
193	-166	3	Clothing	Saree	Clothing Saree	
157	5	9	Clothing	Saree	Clothing Saree	
75	0	7	Clothing	Saree	Clothing Saree	
25	-5	4	Clothing	Saree	Clothing Saree	
43	0	3	Clothing	Saree	Clothing Saree	
160	-59	2	Clothing	Saree	Clothing Saree	
1603	0	9	Clothing	Saree	Clothing Saree	
353	90	8	Clothing	Saree	Clothing Saree	
534	0	3	Clothing	Saree	Clothing Saree	
149	-87	4	Clothing	Saree	Clothing Saree	
635	-349	5	Clothing	Saree	Clothing Saree	

- Calculate Revenue per Order in Order Details Table: Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order

1 Revenue = 'List of Order'[Amount]*'List of Order'[Quantity]							
Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue
B-25602	561	212	3	Clothing	Saree	Clothing Saree	1683
B-25602	119	-5	8	Clothing	Saree	Clothing Saree	952
B-25603	193	-166	3	Clothing	Saree	Clothing Saree	579
B-25604	157	5	9	Clothing	Saree	Clothing Saree	1413
B-25605	75	0	7	Clothing	Saree	Clothing Saree	525
B-25609	25	-5	4	Clothing	Saree	Clothing Saree	100
B-25610	43	0	3	Clothing	Saree	Clothing Saree	129
B-25611	160	-59	2	Clothing	Saree	Clothing Saree	320
B-25613	1603	0	9	Clothing	Saree	Clothing Saree	14427
B-25619	353	90	8	Clothing	Saree	Clothing Saree	2824
B-25622	534	0	3	Clothing	Saree	Clothing Saree	1602
B-25623	149	-87	4	Clothing	Saree	Clothing Saree	596
B-25625	635	-349	5	Clothing	Saree	Clothing Saree	3175
B-25628	24	-9	4	Clothing	Saree	Clothing Saree	96
B-25633	711	-8	4	Clothing	Saree	Clothing Saree	2844
B-25635	382	30	3	Clothing	Saree	Clothing Saree	1146
B-25636	637	113	5	Clothing	Saree	Clothing Saree	3185

Create a Calculated Column to Categorize Sales: Add a calculated column named

'Sales Category' in the Order Details table that categorizes each order as 'Above Average'

or 'Below Average' based on the Amount value.

Sales Category = if('Order Detail'[Amount]>AVERAGE('Order Detail'[Amount]),"Above Average","below Average")							
Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue	Sales Category
561	212	3	Clothing	Saree	Clothing Saree	1683	Above Average
119	-5	8	Clothing	Saree	Clothing Saree	952	below Average
193	-166	3	Clothing	Saree	Clothing Saree	579	below Average
157	5	9	Clothing	Saree	Clothing Saree	1413	below Average
75	0	7	Clothing	Saree	Clothing Saree	525	below Average
25	-5	4	Clothing	Saree	Clothing Saree	100	below Average
43	0	3	Clothing	Saree	Clothing Saree	129	below Average
160	-59	2	Clothing	Saree	Clothing Saree	320	below Average
1603	0	9	Clothing	Saree	Clothing Saree	14427	Above Average
353	90	8	Clothing	Saree	Clothing Saree	2824	Above Average
534	0	3	Clothing	Saree	Clothing Saree	1602	Above Average
149	-87	4	Clothing	Saree	Clothing Saree	596	below Average
635	-349	5	Clothing	Saree	Clothing Saree	3175	Above Average
24	-9	4	Clothing	Saree	Clothing Saree	96	below Average
711	-8	4	Clothing	Saree	Clothing Saree	2844	Above Average
382	30	3	Clothing	Saree	Clothing Saree	1146	Above Average
637	113	5	Clothing	Saree	Clothing Saree	3185	Above Average
122	-47	4	Clothing	Saree	Clothing Saree	488	below Average
20	-8	2	Clothing	Saree	Clothing Saree	40	below Average
42	-6	4	Clothing	Saree	Clothing Saree	168	below Average
55	-26	4	Clothing	Saree	Clothing Saree	220	below Average
130	-41	4	Clothing	Saree	Clothing Saree	520	below Average
211	-105	2	Clothing	Saree	Clothing Saree	422	below Average

Calculated Measures

1. Order Count- Create a measure that calculates the total number of orders in the Order Details table.

Clipboard

2K
Order Count

DAX query view

Structure Formatting Properties

1 Order Count = COUNTROWS('Order Detail')

▼	Amount ▼	Profit ▼	Quantity ▼	Category ▼	Sub-Category ▼	Category Type ▼	Revenue ▼
	561	212	3	Clothing	Saree	Clothing Saree	168
	119	-5	8	Clothing	Saree	Clothing Saree	95
	193	-166	3	Clothing	Saree	Clothing Saree	57
	157	5	9	Clothing	Saree	Clothing Saree	141
	75	0	7	Clothing	Saree	Clothing Saree	52
	25	-5	4	Clothing	Saree	Clothing Saree	10
	43	0	3	Clothing	Saree	Clothing Saree	12
	160	-59	2	Clothing	Saree	Clothing Saree	32
	1603	0	9	Clothing	Saree	Clothing Saree	1442
	353	90	8	Clothing	Saree	Clothing Saree	282
	534	0	3	Clothing	Saree	Clothing Saree	160
	149	-87	4	Clothing	Saree	Clothing Saree	59
	635	-349	5	Clothing	Saree	Clothing Saree	317
	24	-9	4	Clothing	Saree	Clothing Saree	9
	744	0	4	Clothing	Saree	Clothing Saree	224

2. Average Profit in Delhi- Create a measure that calculates the average profit for orders placed in Delhi

43.31
Average Profit in Delhi

Average Profit in Delhi = `Calculate(Average('Order Detail'[Profit]),('List of Orders'[City]="delhi"))`

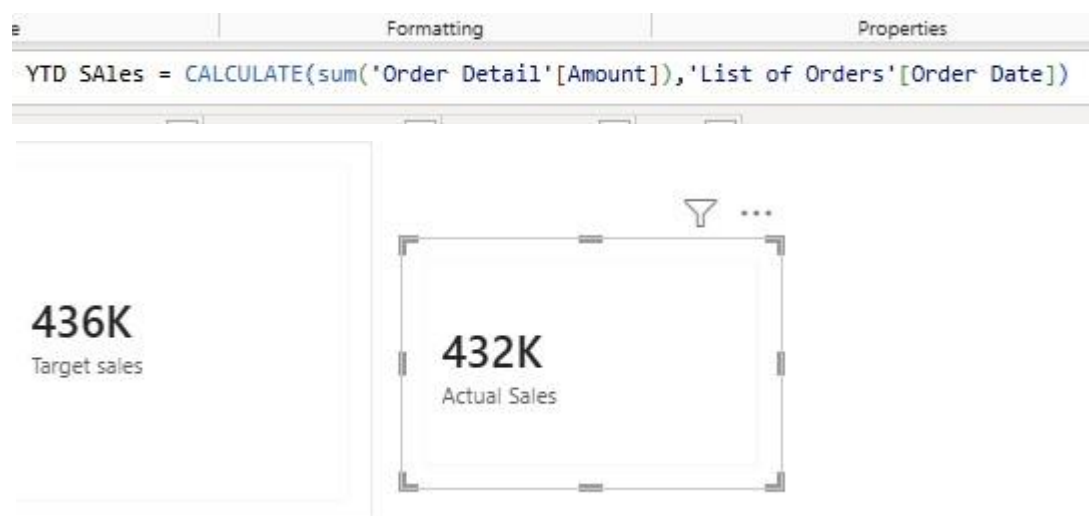
Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue	Sales Category
561	212	3	Clothing	Saree	Clothing Saree	1683	Above Average
119	-5	8	Clothing	Saree	Clothing Saree	952	Below Average
193	-166	3	Clothing	Saree	Clothing Saree	579	Below Average
157	5	9	Clothing	Saree	Clothing Saree	1413	Below Average
75	0	7	Clothing	Saree	Clothing Saree	525	Below Average
25	-5	4	Clothing	Saree	Clothing Saree	100	Below Average
43	0	3	Clothing	Saree	Clothing Saree	129	Below Average
160	-59	2	Clothing	Saree	Clothing Saree	320	Below Average
1603	0	9	Clothing	Saree	Clothing Saree	14427	Above Average
353	90	8	Clothing	Saree	Clothing Saree	2824	Above Average
534	0	3	Clothing	Saree	Clothing Saree	1602	Above Average
149	-87	4	Clothing	Saree	Clothing Saree	596	Below Average
635	-349	5	Clothing	Saree	Clothing Saree	3175	Above Average
24	-9	4	Clothing	Saree	Clothing Saree	96	Below Average
711	-8	4	Clothing	Saree	Clothing Saree	2844	Above Average
382	30	3	Clothing	Saree	Clothing Saree	1146	Above Average
637	113	5	Clothing	Saree	Clothing Saree	3185	Above Average
122	-47	4	Clothing	Saree	Clothing Saree	488	Below Average
20	-8	2	Clothing	Saree	Clothing Saree	40	Below Average
42	-6	4	Clothing	Saree	Clothing Saree	168	Below Average

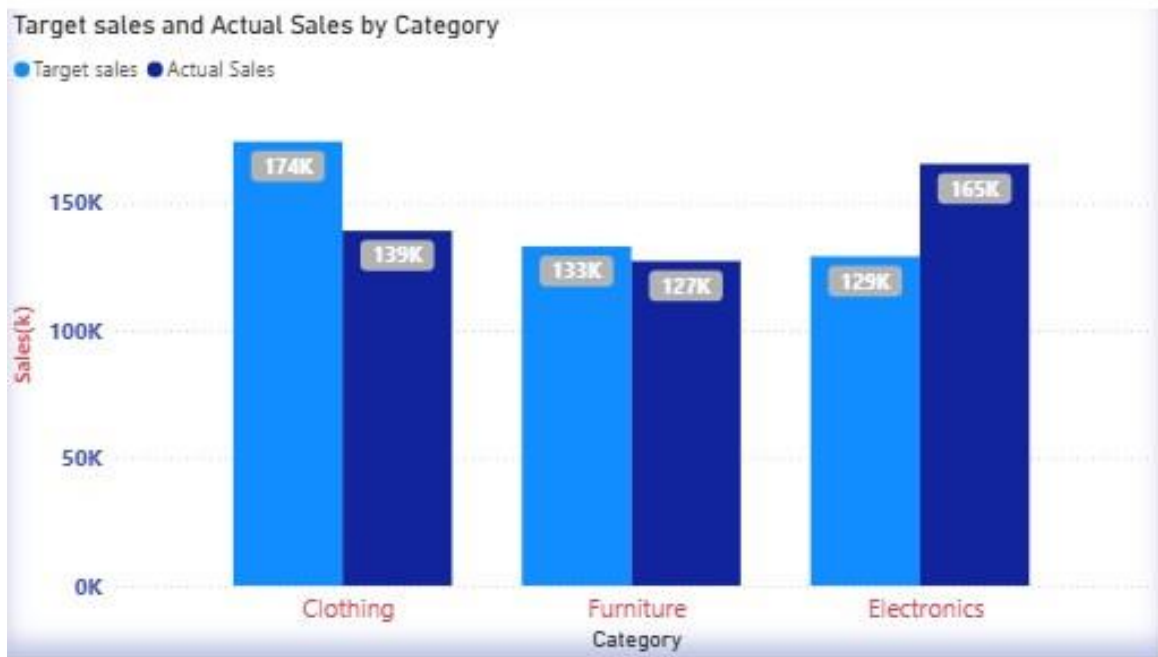
- Year-to-Date (YTD) Sales- Create a measure that calculates total sales accumulated from the beginning of the year up to the current order date.

Data Visualization

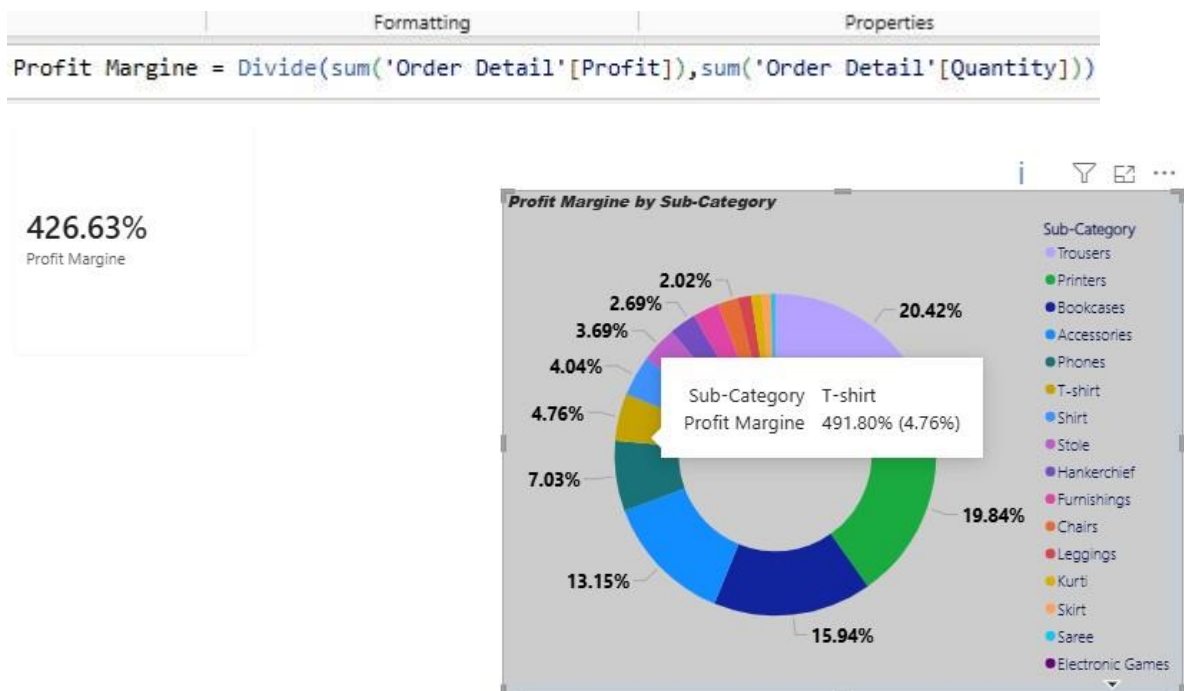
Create the following visualizations using Power BI:

- Sales Target Achievement by Category: Compare actual sales with sales targets by category using a clustered column chart.



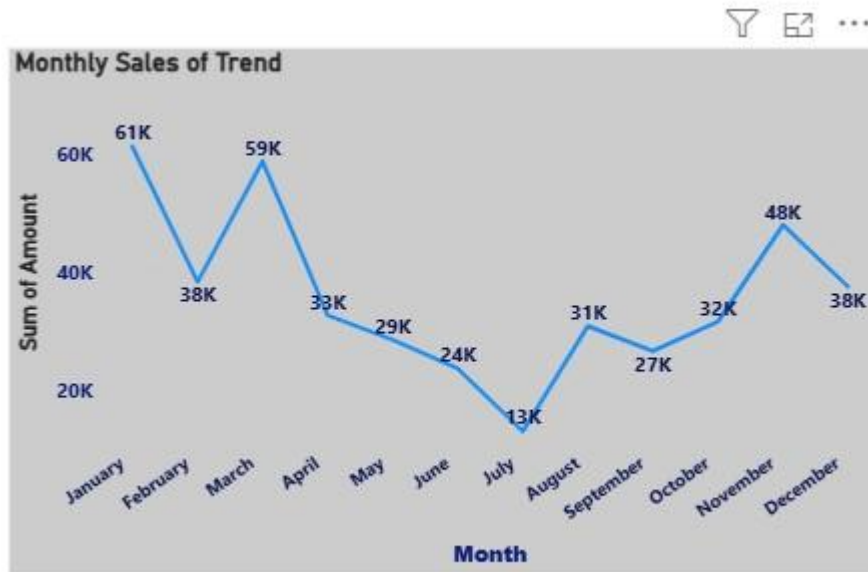


- Max Profit Margin by Sub-Category: Analyse the maximum profit margin for each sub-category of products using a donut chart.

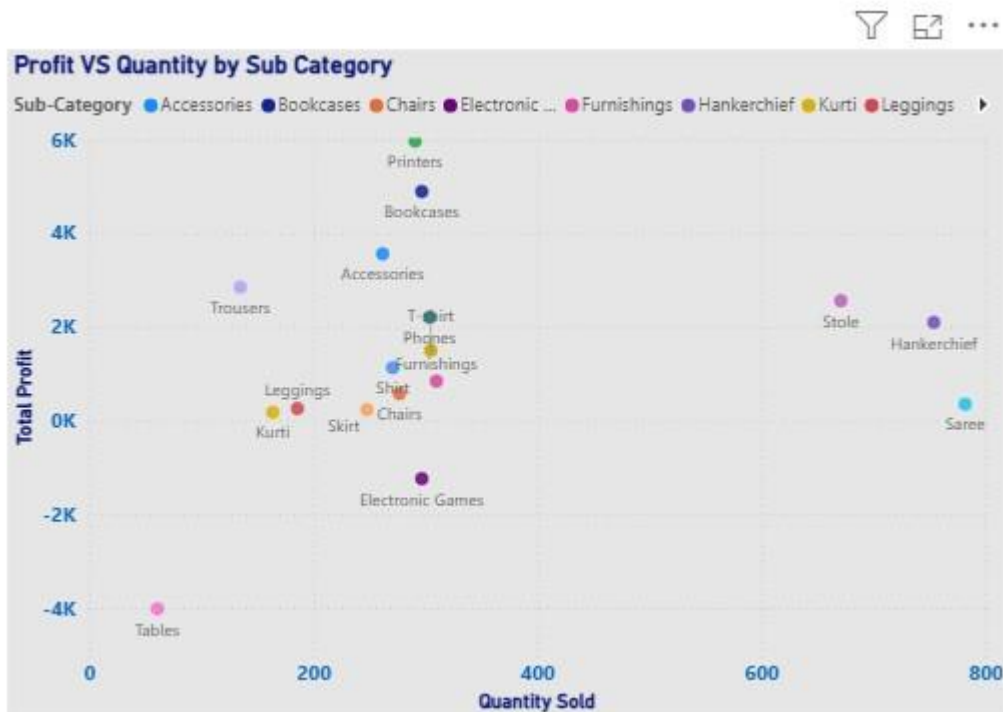


Monthly Sales Trend: Show the trend of monthly sales over time using a line

chart.

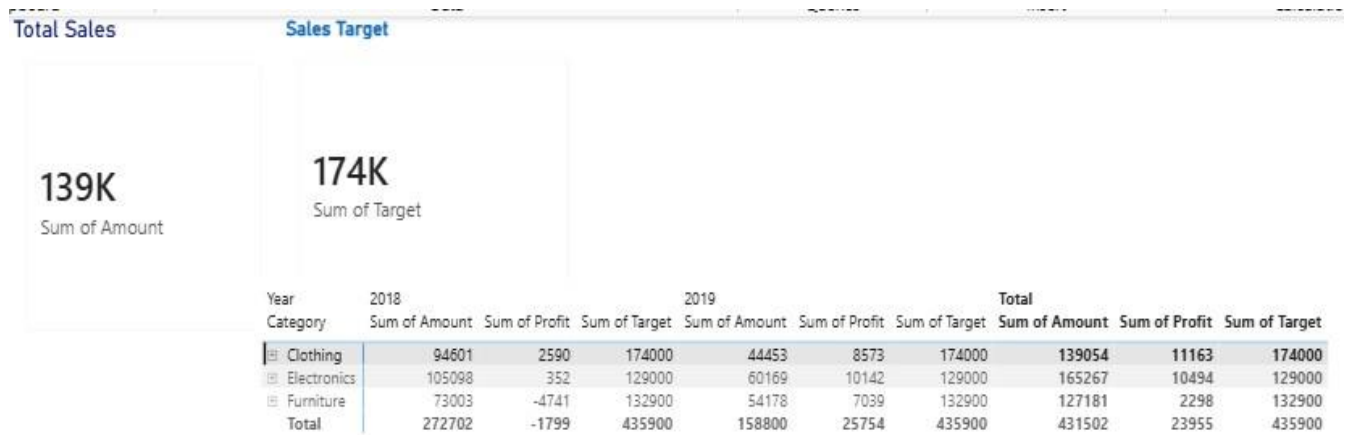


Comparison of Profit and Quantity by Sub-Category: Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.



- Comparison of Total Sales Amount and Target: Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.

- Sales Performance Matrix: Build a matrix view to analyse how actual sales compare to sales targets across different categories and months.



Geographic Sales Analysis: Visualize total sales on a map by city to identify regional sales patterns.



- Sales Distribution by Sub-Category: Represent the sales distribution across different
- sub-categories using a tree



- Order Count Analysis by State: Create a funnel chart to visualize the distribution of order counts across different states.

